- Vectors in underline, matrices in capital. Assume all matrices have real entries unless stated otherwise.
- For a vector $\underline{x}$, we denote by x_i the i^{th} entry of $\underline{x}$.
- The inequality $A \geq 0$ means that the matrix A is positive semi definite. The inequality $\underline{x} \geq 0$ means that all the entries of the vector $\underline{x}$ are non negative.
- Most of these problems are adapted from [1, 2] and related resources on the web.
- You may use either of `CVXPY` (Python), `CVX.jl` (Julia) or `CVX` (matlab) to solve these problems.

1. *Ridge Regression and the Minimum Norm Solution.*

   Consider the ridge regression problem

$$\min_{\underline{x}\in\mathbb{R}^d} \frac{1}{n} \left\|A\underline{x} - \underline{b}\right\|^2 + \lambda \left\|\underline{x}\right\|^2,$$

   where $A \in \mathbb{R}^{n\times d}$, $\underline{b} \in \mathbb{R}^n$, and $\lambda > 0$.

   (a) Derive a closed-form expression for the optimal solution $\underline{x}^*(\lambda)$. Is the solution unique for all n, d, and $\lambda > 0$, regardless of the column rank of A? Justify your answer.

   (b) Now consider the *overparameterized* regime: $n < d$. Compute $\underline{x}^*_{\text{mn}} = \lim_{\lambda\to 0^+} \underline{x}^*(\lambda)$ and simplify your expression as much as possible. Give your answer in terms of the singular value decomposition of A.

   (c) When $\lambda = 0$ and $n < d$, argue why $\min_{\underline{x}} \frac{1}{n} \left\|A\underline{x} - \underline{b}\right\|^2$ fails to have a unique minimizer.

   (d) Despite the non-uniqueness at $\lambda = 0$, the limit $\underline{x}^*_{\text{mn}}$ computed in part (b) exists and is well-defined. Explain why $\underline{x}^*_{\text{mn}}$ is the *minimum-norm* solution to $A\underline{x} = \underline{b}$, i.e., it solves

$$\min_{\underline{x}} \left\|\underline{x}\right\|^2 \quad \text{subject to} \quad A\underline{x} = \underline{b}.$$

2. *Excess Risk of the Minimum-Norm Estimator.*

   Consider the linear model discussed in class: $\underline{y} = A\underline{x}_* + \underline{\varepsilon}$, where $\underline{x}_* \in \mathbb{R}^d$ is a fixed vector, $A \in \mathbb{R}^{n\times d}$ is a fixed matrix, and $\underline{\varepsilon} \sim \mathcal{N}(0, \sigma^2 I_n)$ is noise. You may assume A has thin SVD $A = U\Sigma V^\top$ with singular values $\sigma_1 \geq \cdots \geq \sigma_r > 0$ ($r = \text{rank}(A)$).

   (a) Using the expression for the risk (testing error) $\frac{1}{n}\mathbb{E}\big[\big\|A\hat{\underline{x}}(\lambda) - \underline{y}\big\|^2\big]$ of the ridge estimator derived in class, write the bias and variance terms explicitly as functions of λ and $\{\sigma_i\}$.

   (b) Take $\lambda \to 0^+$ to derive the limiting risk (testing error)

$$\frac{1}{n}\mathbb{E}\big[\big\|A\hat{\underline{x}}_{\text{mn}} - \underline{y}\big\|^2\big].$$

   Express your answer in terms of n, r, and σ^2.

3. *Parameter Estimation Error of the Minimum-Norm Estimator.* This question follows from the previous two questions.

   Consider the linear model discussed, and assume the overparameterized regime $n < d$. Derive the parameter estimation error $\frac{1}{n}\mathbb{E}\big[\|\hat{\underline{x}}_{\text{mn}} - \underline{x}_*\|^2\big]$. Here $\hat{\underline{x}}_{\text{mn}}$ is the minimum-norm solution, as in Problem 1. This analysis will follow similar to the risk analysis done in class.

4. *Efficient Cross-Validation [3, Ex. 13.21].*

   The cost of fitting a model with p basis functions and N data points (using QR factorization) is $2Np^2$ flops. We explore the complexity of 10-fold cross-validation on a fixed dataset. Divide the data into 10 folds of $N/10$ points each. The naïve method fits 10 separate models, each on 9 folds, costing

$10 \cdot 2(0.9N)p^2 = 18Np^2$ flops—roughly $10\times$ the cost of fitting a single model. (Evaluating each model on its held-out fold costs $2(N/10)p$ flops, which is negligible.)

The following method achieves the same result more efficiently. Let $A_1, \ldots, A_{10} \in \mathbb{R}^{(N/10)\times p}$ and $b_1, \ldots, b_{10} \in \mathbb{R}^{N/10}$ denote the data matrix blocks and right-hand-side vectors for each fold.

(a) Form the Gram matrices $G_i = A_i^\top A_i$ and the vectors $c_i = A_i^\top b_i$ for $i = 1, \ldots, 10$. What is the total flop count for this step?

(b) Form the full-data aggregates $G = \sum_{i=1}^{10} G_i$ and $c = \sum_{i=1}^{10} c_i$. What is the flop count?

(c) For each fold $k = 1, \ldots, 10$, compute the leave-one-fold-out estimator

$$\hat{\theta}_k = (G - G_k)^{-1}(c - c_k).$$

What is the flop count per fold and in total across all 10 folds?

(d) Sum the flop counts from parts (a)–(c) and compare the total to the naïve cost of $18Np^2$. For what regime of N and p is the efficient method advantageous?

5. *Polynomial Regression: Ordinary and Ridge (Coding).*

We consider the problem of fitting a polynomial of degree at most $d-1$ to n data points , where $(x_i, y_i) \in \mathbb{R}^2$. The data input for this problem consists of points $\{(x_i, y_i)\}_{i=1}^n$.

(a) *Problem setup.* Out goal is to find the polynomial $p(x) = c_0 + c_1 x + \cdots + c_{d-1}x^{d-1}$, where $\underline{c} \in \mathbb{R}^d$, such that the objective

$$\frac{1}{n}\sum_{i=1}^{n}(y_i - p(x_i))^2$$

is minimized. Rewrite the objective as $\left\| A\underline{c} - \underline{y} \right\|^2$. Can you comment on the structure of the matrix A?

(b) *Dataset generation.* Using the companion notebook `A4_polynomial_regression.ipynb`, generate a dataset as follows. Draw $x_1, \ldots, x_n \overset{\text{iid}}{\sim} \text{Uniform}[-1, 1]$, and set

$$y_i = x_i^2 - \tfrac{1}{2} + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2).$$

Fix $\sigma = 0.2$ and $d = 6$. Complete the skeleton functions in the notebook. You may use the provided code to generate the dataset and implement the OLS and ridge regression solvers, or you may implement your own.

(c) *Ordinary least squares (OLS).* Fix $d = 6$. For each value of n in a suitable range:

(i) Randomly split the data into a training set of size n and a held-out test set of size $n_{\text{test}} = 500$.

(ii) Solve the OLS problem using the training set to obtain $\hat{\underline{c}}$.

(iii) Evaluate the test risk $R_{\text{test}} = \frac{1}{n_{\text{test}}} \left\| A_{\text{test}}\hat{\underline{c}} - \underline{y}_{\text{test}} \right\|^2$.

(iv) Repeat for $T = 100$ independent trials and report the average test risk $\bar{R}(n)$.

How is this setup different from the one discussed in class when we derived the formula for the expected test risk? Nevertheless, does the empirical $\bar{R}(n)$ agree with the formula for the expected test risk derived in class?

(d) *Ridge regression.* Repeat part (c) using ridge regression with regularization parameter $\lambda > 0$. At each value of n, select λ by 5-fold cross-validation on the training set over a logarithmically spaced grid; a complete implementation of cross-validation is provided in the notebook. You need only implement the `ridge` solver. Report the average test risk $\bar{R}_{\text{ridge}}(n)$.

(e) *Risk vs. sample size.* On a single plot, display $\bar{R}(n)$ (OLS) and $\bar{R}_{\text{ridge}}(n)$ (ridge) as functions of n, with $d = 6$ fixed. Describe the qualitative behavior of each curve and comment on the effect of regularization.

References

[1] Stephen Boyd and Lieven Vandenberghe. *Convex optimization*. Cambridge university press, 2004.

[2] Stephen Boyd and Lieven Vandenberghe. Additional exercises for convex optimization, 2010.

[3] Stephen Boyd and Lieven Vandenberghe. *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares*. Cambridge University Press, 2018.